Calcurion Maths Worksheets

A-Level Further Mathematics

Topic: Further Pure 1 – Further Vectors

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. Find $\mathbf{a} \times \mathbf{b}$ for $\mathbf{a} = \mathbf{i} + 2\mathbf{j} - \mathbf{k}$, $\mathbf{b} = 2\mathbf{i} - \mathbf{j} + 3\mathbf{k}$. **[4]**

2. Find the area of the parallelogram formed by $\mathbf{a}$ and $\mathbf{b}$ above. **[2]**

3. Find a unit vector perpendicular to both $\mathbf{i} + \mathbf{j}$ and $\mathbf{j} + \mathbf{k}$. **[4]**

4. Find the volume of the parallelepiped formed by $\mathbf{a} = \mathbf{i}$, $\mathbf{b} = \mathbf{j}$, $\mathbf{c} = \mathbf{i} + \mathbf{j} + \mathbf{k}$. **[3]**

5. Find the equation of the plane through $A(1, 0, 0)$, $B(0, 1, 0)$, $C(0, 0, 1)$ in vector form. **[4]**

6. Find the Cartesian equation of the plane $\mathbf{r} \cdot (2\mathbf{i} - \mathbf{j} + \mathbf{k}) = 5$. **[2]**

7. Find the angle between the planes $x + y + z = 1$ and $2x - y + z = 3$. **[4]**

8. Find the distance from $(1, 2, 3)$ to the plane $x + y + z = 6$. **[3]**

9. Find the line of intersection of the planes $x + y + z = 1$ and $x - y + z = 2$. **[5]**

10. Determine whether the lines $\mathbf{r} = \mathbf{i} + t\mathbf{j}$ and $\mathbf{r} = \mathbf{j} + s\mathbf{k}$ are skew. **[3]**

11. Find the shortest distance between the skew lines $\mathbf{r} = \mathbf{i} + t(\mathbf{j} + \mathbf{k})$ and $\mathbf{r} = \mathbf{i} + \mathbf{j} + s(\mathbf{i} - \mathbf{k})$. **[5]**

12. Find the foot of the perpendicular from the origin to the plane $x + 2y + 2z = 9$. **[4]**

13. Show that $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$ is the scalar triple product and state its geometric meaning. **[3]**

14. Find $\mathbf{a} \cdot (\mathbf{a} \times \mathbf{b})$. **[2]**

15. Find a vector equation of the plane containing the line $\mathbf{r} = \mathbf{i} + t(\mathbf{j} - \mathbf{k})$ and the point $(1, 1, 1)$. **[4]**

Section B – Hard (Q16–25)

16. Find the equation of the plane bisecting the line segment joining $A(1, 2, 3)$ and $B(3, 4, 5)$ perpendicularly. **[4]**

17. Find the reflection of the point $(1, 0, 0)$ in the plane $x + y + z = 3$. **[5]**

18. Find the angle between the line $\mathbf{r} = \mathbf{i} + t(\mathbf{i} + \mathbf{j} + \mathbf{k})$ and the plane $x + y + z = 1$. **[4]**

19. Determine the values of λ for which the lines $\mathbf{r} = \mathbf{i} + \lambda\mathbf{j} + t(\mathbf{j} + \mathbf{k})$ and $\mathbf{r} = \mathbf{j} + s(\mathbf{i} + \mathbf{k})$ intersect. **[5]**

20. Find the volume of the tetrahedron with vertices at O, $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$. **[3]**

21. Prove that $(\mathbf{a} \times \mathbf{b}) \cdot (\mathbf{c} \times \mathbf{d}) = (\mathbf{a} \cdot \mathbf{c})(\mathbf{b} \cdot \mathbf{d}) - (\mathbf{a} \cdot \mathbf{d})(\mathbf{b} \cdot \mathbf{c})$. **[5]**

22. Find the Cartesian equations of the line of intersection of $x + y = 1$ and $y + z = 2$. **[4]**

23. A plane passes through $(1, 0, 0)$ and contains the line $\mathbf{r} = t(\mathbf{i} + \mathbf{j})$. Find its equation. **[4]**

24. Find the distance between the parallel planes $x + y + z = 1$ and $x + y + z = 5$. **[3]**

25. Find the point on the plane $x + 2y - z = 4$ closest to $(1, 1, 1)$. **[5]**

End of Worksheet • *Show all working clearly.*